EE 3072.1/372.1
Dynamic Systems and Control
Final Exam
June 14, 2022
Time Limit: 75 Minutes

Student Name: _____

Student Number: _____

You may *not* use your books, notes on this exam. You are allowed to use a non-programmable calculator and an A4-sized paper with handwritten formulas on both sides. You are NOT allowed to write any examples or solutions.

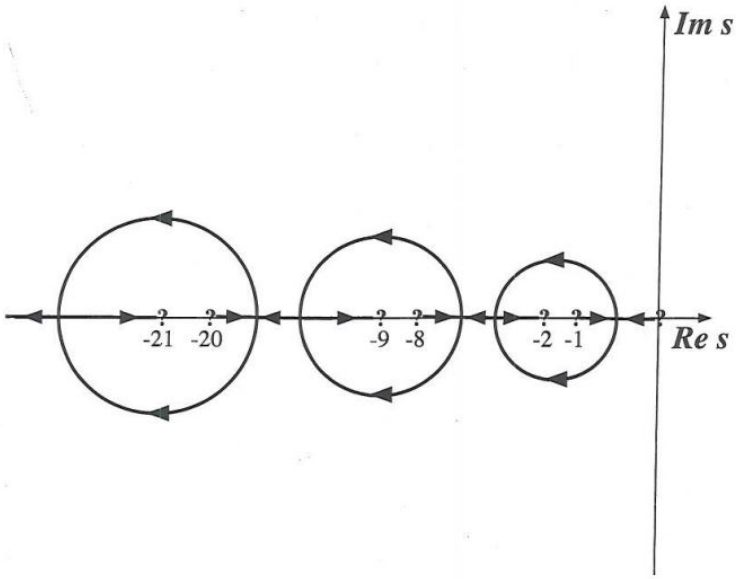
An incorrect answer will receive no credit. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit.

Question:	1	2	3	4	Total
Points:	10	20	30	40	100
Score:					

Questions

1. (10 points) The root locus of a system $G(s)$ looks like

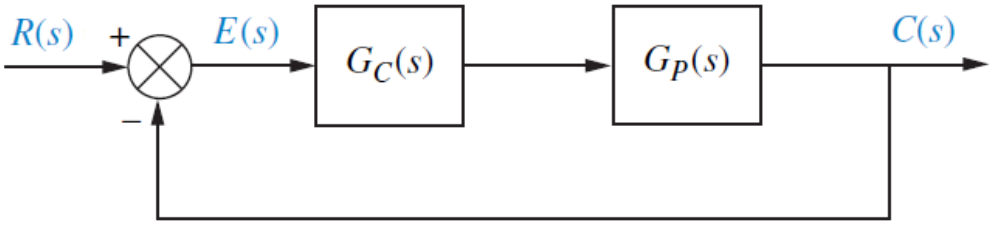
   where the symbols '?' represent open-loop poles/zeros. Write the expression for $G(s)$. Note that you must justify your answer to get credits.



Answer: $G(s) = \dfrac{(s+2)(s+9)(s+21)}{s(s+1)(s+8)(s+20)}$

Justification: Root locus starts at open loop poles and terminates at open loop zeros.

2. (20 points) Consider the feedback control of the plant $G_P(s) = \dfrac{1}{s(s+2)(s+10)}$. Design a proportional-derivative controller $G_C(s) = K_P + K_D s$ for which the settling time of the system is 4 seconds (2% error band) and the maximum overshoot is 10% for a unit step input.



Answer:

For a settling time of 4 seconds, we need $4/\zeta\omega_n = 4$ and for a maximum overshoot of 10%, we need $\zeta = 0.6$ which yields $\omega_n = 5/3$. The second order system corresponding to these values is $s^2 + 2s + 25/9$.

The characteristic polynomial of the system can be computed as

$$s(s+2)(s+10) + K_P + K_D s = s^3 + 12s^2 + (20 + K_D)s + K_P$$

Let $-a$ $(a > 0)$ denote the third closed loop pole (that is not dominant). Then we need

$$s^3 + 12s^2 + (20 + K_D)s + K_P = (s + a)(s^2 + 2s + 25/9)$$

which yields $a = 10$, $K_P = 250/9$ and $K_D = 25/9$. Note that the real parts of the dominant poles are equal to -1 whereas the non-dominant pole is located at -10.

3. (30 points) Consider the feedback control of the plant $G_P(s) = \dfrac{1}{(s+3)(s+2)(s+a)}$ with a PI controller $G_C(s) = K_p + \dfrac{K_I}{s}$.

   a) (15 pts) Find the ranges for the controller gains K_P and K_I (as functions of the constant a) so that the closed-loop system is stable.

   b) (15 pts) Let $K_I = K_p$ and $a = 1$. Sketch the locations of the closed loop poles with respect to K_P.

   Answer:

   a) The characteristic equation of the closed-loop system is $1 + G_P(s)G_C(s) = 0$ which can be rewritten as

$$s^4 + (a+5)s^3 + (5a+6)s^2 + (6a+K_P)s + K_I = 0$$

   The corresponding Routh array is

$$\begin{array}{c|ccc} s^4 & 1 & 5a+6 & K_I \\ s^3 & a+5 & 6a+K_P & 0 \\ s^2 & 5a+6-\dfrac{6a+K_P}{a+5} & K_I & \\ s^1 & 6a+K_P-\dfrac{(a+5)K_I}{5a+6-\frac{6a+K_P}{a+5}} & 0 & \\ s^0 & K_I & & \end{array}$$

   For stability, we need all values in the first column to be positive, i.e.,

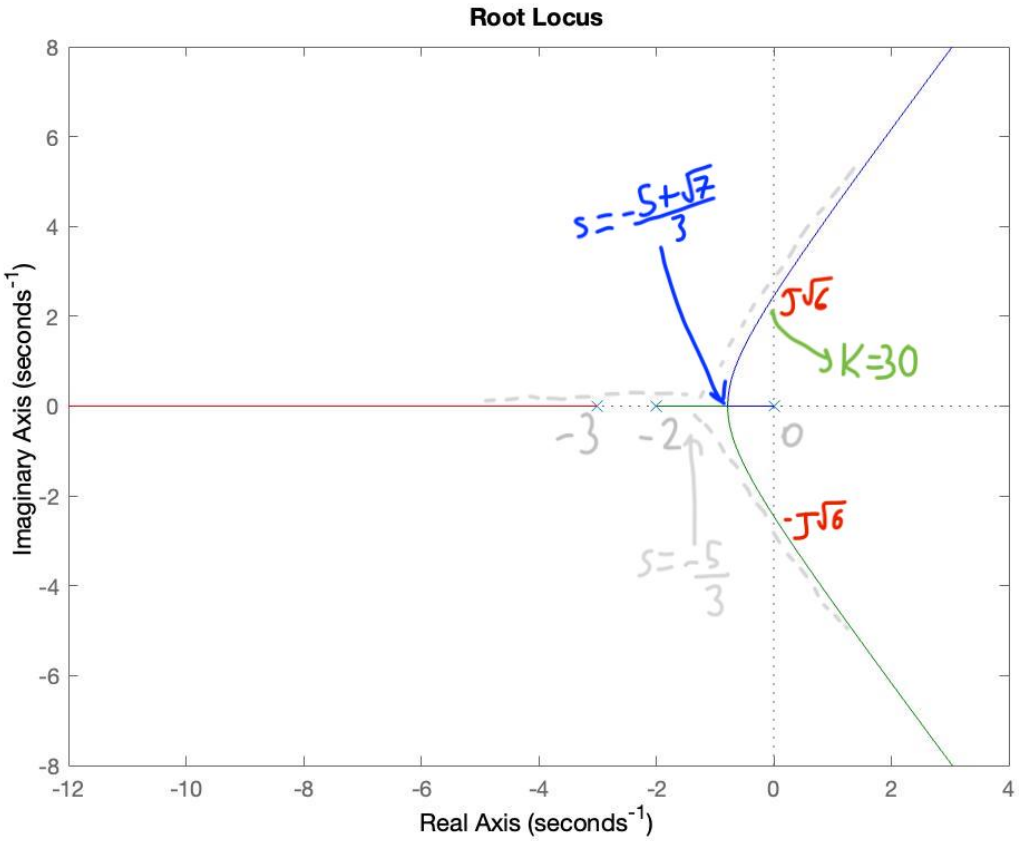
$$K_P < 5(a^2+5a+6)$$

$$0 < K_I < \frac{(6a+K_P)[5(a^2+5a+6)-K_P]}{(a^5)^2}$$

   b) For $K_I = K_P$ and $a = 1$, the loop transfer function can be computed as $G_P(s)G_C(s) = \dfrac{K_P}{s(s+2)(s+3)}$. There are three branches starting at $s = -3$, $s = -2$ and $s = 0$. The real-axis segments $s \le -3$ and $-2 \le s \le 0$ are parts of the locus. There are 3 asymptotes centered at $\frac{-3-2}{3} = -\frac{5}{3}$ and the angles are $\pm 60^\circ$ and 180°.

   The breakaway point can be computed from $\dfrac{\partial K}{\partial s} = 0 \implies 3s^2 + 10s + 6 = 0$ which has the real root $s_{ba} = \dfrac{-5+\sqrt{7}}{3}$ included in the root locus.

   For stability of the closed loop system, we need the roots of the characteristic equation $s^3 + 5s^2 + 6s + K_P = 0$ to be in LHP. Applying Routh's stability test we conclude that $K_P = 30$ at the margin of stability. Furthermore, for $K_P = 30$, the roots are at $\pm j\sqrt{6}$. Therefore, the root locus can be sketched as shown in the figure below.


Root Locus
Imaginary Axis (seconds^{-1})
Real Axis (seconds^{-1})
8
6
4
2
0
-2
-4
-6
-8
-12
-10
-8
-6
-4
-2
0
2
4
$s=-\frac{5+\sqrt{7}}{3}$
$j\sqrt{6}$
K=30
-3
-2
0
$-j\sqrt{6}$
$s=-\frac{5}{3}$

4. (40 points) Consider the system

$$G(s) = \frac{K}{s(s+1)(s+3)}$$

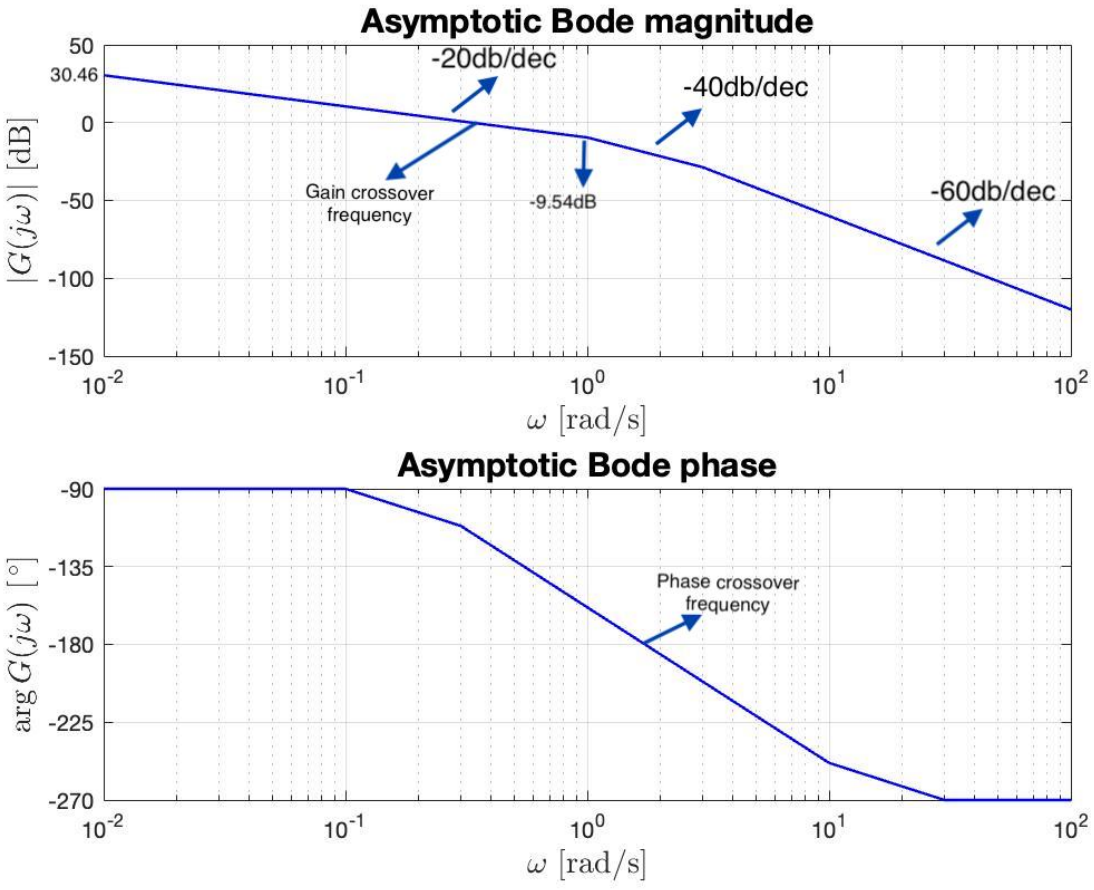
in the negative unity feedback loop.

a) (15 pts) For $K = 1$, sketch the Bode (magnitude and phase) plot of $G(s)$. Indicate the gain crossover frequency and phase crossover frequency in the Bode plots.

b) (20 pts) For $K = 1$, find the phase crossover frequency, gain crossover frequency, phase margin and gain margin, analytically. (In case you need, the roots of the equation $x^3 + 10x^2 + 9x - 1 = 0$ can be computed as -8.9861, -1.1138 and 0.0999.)

c) (5 pts) Sketch the Nyquist diagram of $G(s)$.

Answer:



Gain crossover frequency is the frequency for which $|G(j\omega)| = 0dB$ and phase crossover frequency is the frequency for which $\angle G(j\omega) = -180°$

b)

$$G(j\omega) = \frac{K}{j\omega(j\omega+1)(j\omega+3)} = \frac{-Kj}{\omega(3-\omega^2+4j\omega)}$$
$$= \frac{-Kj(3-\omega^2-4j\omega)}{\omega((3-\omega^2)^2+16\omega^2)} = \frac{-4K\omega + jK(\omega^2-3)}{\omega((3-\omega^2)^2+16\omega^2)}$$

At the phase crossover frequency ω_p, we have $Im(G(j\omega_p)) = 0 \implies \omega_p = \sqrt{3}$. At the phase crossover frequency, $G(j\omega_p) = -1/12$ and therefore gain margin is 12 or $20log(12) = 21.58dB$.

At the gain crossover frequency ω_g, we have $|G(j\omega_g| = 1$ For $K = 1$, we need $|-4\omega_g + j(\omega_g^2 - 3)| = |\omega_g((3-\omega_g^2)^2 + 16\omega_g^2)|$ which can be solved for $\omega_g = 0.3161$. At the gain crossover frequency we have $G(j\omega_g) = -0.3996 - j0.9167$ which has a phase of $-113.55°$. Therefore, the phase margin is $180° - 113.55° = 66.45°$.

c) Note that $Re(G(j0)) = -\frac{4}{9}$ and $Im(G(j0)) = -\infty$. Note also that $Re(G(j\infty)) = -0$ and $Im(G(j\infty)) = +0$. Intersection of the Nyquist plot with the real axis can be computed from $Im(G(j\omega)) = 0 \implies \omega_1 = \sqrt{3} \implies Re(G(j\omega_1)) = -\frac{1}{12}$ From the above calculation, Nyquist plot can be roughly plotted as follows:

